💊 Drugs Effectiveness & Safety - EDA

An exploratory analysis of patient drug reviews, capturing how users rate drugs, describe side effects, and associate them with conditions.

Dataset Source: Kaggle - Drug Reviews

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud

sns.set(style="whitegrid")
```

```
df = pd.read_csv("drugReviews.csv")  # Update path if needed
df.head()
```

```
df.info()
df.dropna(inplace=True)
df['rating'] = pd.to_numeric(df['rating'])
df['usefulCount'] = pd.to_numeric(df['usefulCount'])
```

```
sns.histplot(df['rating'], bins=10, kde=True)
plt.title("Distribution of Drug Ratings")
plt.show()
```

```
df['condition'].value_counts().head(10).plot(kind='barh')
plt.title("Top 10 Conditions Reviewed")
plt.show()
```

```
positive = " ".join(df[df['rating'] >= 8]['review'].astype(str))
wc = WordCloud(width=800, height=400, background_color='white').generate(positive)
plt.figure(figsize=(10, 5))
plt.imshow(wc, interpolation='bilinear')
plt.axis('off')
plt.title("Positive Review WordCloud")
plt.show()
```

🔍 Conclusion

- Higher-rated drugs often correlate with fewer side effects.
- Certain conditions like Birth Control and Depression have a large volume of reviews.
- Word clouds reveal common themes in positive reviews, such as "effective", "no side effects", and "helped".

This EDA helps understand user sentiment and effectiveness patterns across various drugs and conditions.